Addition of pyroligneous extract (wood vinegar) to the diet of growing and finishing pigs

Adição de extrato pirolenhoso à dieta de suínos em crescimento e terminação

Adición de extracto pirolignoso a la dieta de cerdos en crecimiento y acabado

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ABSTRACT
The search for growth promoters from natural resources is essential for swine breeding. This study evaluated the effect of including different levels of eucalyptus wood vinegar (WV) in the swine diet on the performance, quantitative and qualitative characteristics of the carcass, and diarrhea index. Twenty-four crossbred females (Pietrain, Landrace, and Large White lines) were used, with an average initial weight of 31.2 ± 5.6 kg. The animals received feed and water *ad libitum*. The feed was composed of corn, soybean meal, vegetable oil, and commercial supplement (evimix), and the dietary experiments consisted of rising levels of pyroligneous extract (0, 0.5, 1.0, and 1.5%). At the end of the experimental period (72 days), when the animals had an average weight of 97.5 kg, the following performance parameters were measured: daily feed consumption (DFC), daily weight gain (DWG), food recovery (FR), and quantitative and qualitative carcass limits. The results demonstrated no linear or quadratic effect ($P<0.05$) on the growth phase between treatments. However, considering the total period (growth phase + termination) of the animals in the 1% group, they presented a linear effect ($P<0.05$) for DFC but no significant effect for DWG and FR. There was no difference ($P<0.05$) between quantitative and qualitative carcass values. Evaluation of the diarrhea index showed that the WV did not change like the animals' feces. So, WV can be used up to 1.5% of the diet of these animals without causing damage to performance or carcass quality, in addition to shortening the time to slaughter by 10%.

Keywords: growth promoters, performance, eucalyptus wood vinegar

RESUMO
Objetivou-se com este estudo avaliar os efeitos de diferentes níveis de inclusão do extrato pirolenhoso (EP) sobre o desempenho, características de carcaça, qualidade de carne e índice de diarreia em suínos nas fases de crescimento e terminação. Foram utilizadas 24 fêmeas mestiças (Pietrain, Landrace e Large White), com peso médio inicial de 31,2+ 5,6 kg. Os animais foram alojados em um galpão experimental com comedouros semiautomáticos e bebedouros tipo chupeta. As dietas experimentais foram compostas por milho, farelo de soja, óleo vegetal e núcleo comercial, sendo suplementadas com níveis crescentes de inclusão do extrato pirolenhoso (0; 0,5; 1,0 e 1,5%). Os animais foram pesados a cada 14 dias para o estudo dos parâmetros de desempenho: consumo de ração diário (CRD), ganho de peso diário (GPD), e conversão alimentar (CA). No final eles foram abatidos e os parâmetros quantitativos e qualitativos das carcaças coletados. Foi observado efeito significativo ($P<0,05$) no consumo de ração diário e no ganho de peso diário. Quanto as variáveis de carcaça e qualidade de carne, não se observou efeito ($P>0,05$) dos níveis testados. O extrato pirolenhoso pode ser utilizado até o nível de 1,5% nas dietas de suínos em crescimento e terminação, pois promove aumento do consumo de ração diário e do ganho de peso diário, sem causar prejuízos nas características quantitativas ou qualitativas de carcaça, além de antecipar o abate.

Palavras-chave: aditivos antimicrobianos, promotores de crescimento, vinagre de madeira, vinagre de bambu.

RESUMEN
El objetivo de este estudio fue evaluar los efectos de diferentes niveles de inclusión de extracto piroligneo (PE) sobre el rendimiento, las características de la canal, la calidad de la carne y las tasas de diarrea en cerdos durante las fases de crecimiento y acabado. Se utilizaron 24 cerdas cruzadas (Pietrain, Landrace y Large White), con un peso inicial medio de 31,2+5,6 kg. Los animales se alojaron en una nave experimental con comederos semiautomáticos y bebederos tipo chupete. Las dietas experimentales consistieron en maíz, harina de soja, aceite vegetal y núcleo comercial, suplementados con niveles crecientes de extracto pirolignoso (0, 0,5, 1,0 y 1,5%). Los animales se pesaron cada 14 días para estudiar los parámetros de rendimiento: consumo diario de pienso (CRD), ganancia diaria de peso (GPD) y conversión alimenticia (FC). Al final se sacrificaron y se recogieron los parámetros cuantitativos y cualitativos de las canales. Se observó un efecto significativo ($P<0,05$) sobre el consumo diario de pienso y la ganancia diaria de peso. En cuanto a las variables de calidad de la canal y la carne, no hubo efecto ($P>0,05$) de los niveles ensayados. El extracto piroligneo puede utilizarse hasta un nivel del 1,5 por ciento en las dietas de cerdos en crecimiento y acabado, ya que promueve un aumento del consumo diario de pienso y de la ganancia diaria de peso, sin causar ningún perjuicio a las características cuantitativas o cualitativas de la canal, además de adelantar el sacrificio.

Palabras clave: aditivos antimicrobianos, promotores del crecimiento, vinagre de madera, vinagre de bambú.

1 INTRODUCTION

Brazil is the largest producer of charcoal from planted forests, mainly with eucalyptus species, due to its favorable edaphoclimatic conditions and extensive territorial range. In this context, the country has great potential for the production of eucalyptus wood vinegar, a product that can generate extra income for the agricultural sector, as a natural antifungal and antibiotic substance (Baimark & Naimsa, 2009; Bortoletto Et Al., 2009), growth stimulator (Schnitzer Et Al., 2015), and to control pests and diseases (Zanetti Et Al., 2003). The antimicrobial properties of WV also enable its use as an input for animal production (Qu et al., 2018).

Antimicrobial feed additives benefit pig farming significantly, with proven efficacy observed over many decades of use (Gaviolli Et Al., 2013). However, its use has been questioned by the scientific community since indiscriminate inclusion can create resistant bacterial strains and leave residues in the meat that will be consumed (Santos et al., 2016; Lopes et al., 2016). In this context, the use of antimicrobial agents as growth promoters is restricted in many countries (MORAIS et al., 2023). To solve the problem, some alternative products have already been studied, such as probiotics, prebiotics, and herbal medicines (Na et al., 2008), with generally good results.

Another alternative that has been studied is the use of wood vinegar (WV), also known as pyroligneous extract vinegar, obtained through the condensation of smoke generated in the production of charcoal. This product contains substances that can benefit livestock, such as syringol, guaiacol, phenols, furfural, eugenol, and acetic acid, which can act as natural antimicrobials

(Pimenta et al., 2018). Feijó et al. (2022) evaluated EP as an antiseptic in post-surgical procedures in cats and observed inhibition of the action of gram-positive and gram-negative bacteria. Chu et al. (2013) analyzed the performance, immune response, and population of the fecal microflora of growing pigs fed with charcoal diets with bamboo WV as alternatives to antibiotics. They observed that using only 0.3% charcoal or bamboo WV improved the fecal microflora, performance, and feed efficiency of the animals.

Furthermore, Yan et al. (2016) studied the influence of bamboo vinegar supplementation at different levels of inclusion (0, 0.1, and 0.2%) in the diets of finishing pigs on performance, digestibility, meat quality, gas content, harmful fecal matter and fecal microbial concentration They concluded that bamboo vinegar provided beneficial effects on performance and nutrient digestibility through its impact on intestinal microflora, in addition to positively affecting meat quality and reducing the concentration of harmful fecal gases (NH_3 and H_2S). In a study comparing the effect of antibiotics, organic acids, and pyroligneous extract in weanling pig diets, Choi et al. (2009) concluded that although the antibiotic diet had favorable results for performance, animals that were fed diets containing WV had better results than those given organic acids, suggesting that the product can improve the digestibility of nutrients and selectively inhibit pathogenic coliforms.

In this context, the use of pyroligneous extract can benefit animal health, improve the performance and quality of meat, and allow the use of residue from charcoal production.

2 MATERIAL AND METHODS

The Committee on Ethics in Animal Experimentation of Rio Grande do Norte Federal University (UFRN) – protocol number 215.002/2020 approved all the experimental protocols and procedures. The experiment was carried out in the municipality of Macaíba, Rio Grande do Norte State (RN) (5° 53' S, 35° 23' W, and 40 m altitude). Twenty-four female crossbred pigs (Pietrain x Landrace x Large White), weighing 31.2+5.6 kg, were housed in a barn equipped with semi-automatic feeders and waterers.

The animals were distributed in a randomized block design as a function of weight, with four experimental treatments and six replicates lasting 72 days. The average, minimum, and maximum temperatures recorded during the experimental period (72 days) were 28.3, 22.9, and 33.8 °C, and relative humidity was 50.8%.

The experimental diets comprised cornmeal, soybean meal, and a commercial supplement (evimix). The feed was formulated according to the recommendations of Rostagno et al. (2017) and supplemented with rising levels of eucalyptus WV (0, 0.5, 1.0, and 1.5% based on feed weight) for the growth and finishing phases (Tables 1 and 2).

Table 1. Nutritional composition of the feed provided to the hogs in the growing stage

Ingredients (%)	Levels of WV (%)			
	0.0	0.5	1.0	1.5
Cornmeal 7.88%	69.69	69.69	69.69	69.69
Soybean meal 45 %	23.71	23.71	23.71	23.71
Commercial supplement (evimix)[1]	3.00	3.00	3.00	3,00
Soy oil	1.68	1.68	1.68	1.68
L-Lysine[2]	0.280	0.280	0.280	0.280
L-Threonine[2]	0.078	0.078	0.078	0.078
DL-Methionine[2]	0.054	0.054	0.054	0.054
Wood vinegar	0	0.5	1.0	1.5
Inert material	1.5	1.0	0.5	0
Total (kg)	100.00	100.00	100.00	100.00
Calculated composition				
Metabolizable energy (kcal kg^{-1})	3,230	3,230	3,230	3,230
Crude protein (%)	16.57	16.57	16.57	16.57
Digestible lysine (%)	0.945	0.945	0.945	0.945
Digestible methionine (%)	0.284	0.284	0.284	0.284
Digestible threonine (%)	0.620	0.620	0.620	0.620
Digestible tryptophan (%)	0.170	0.170	0.170	0.170

[1]Guaranteed levels per kg of product: folic acid (min) 5 mg; panthotenic acid (min) 257 mg; BHT (min) 133 mg; bacitracin (min) 1.333.33 mg; biotin (min) 3.0 mg; calcium (min) 230 g; calcium (max) 240 g; cobalt (min) 5.66 mg; copper (min) 4,000 mg; choline (min) 3,330 mg; iron (min) 3,333 mg; phytase (min) 16.65 ftu; fluorine (max) 332 mg; phosphorous (min) 34.6 g; iodine (min) 33.33 mg; manganese (min) 1,333 mg; niacin (min)569 mg; selenium (min) 10 mg; sodium (min) 58.5 g; vitamin A (min) 116,800 UI; vitamin B1 (min) 16.7 mg; vitamin B12 (min) 353 mg; vitamin B2 (min) 66.7 mg; vitamin B6 (min) 16.7 mg; vitamin D3 (min) 25,000 UI; vitamin C (min) 833.33 UI; vitamin K3 (min) 49 mg; zinc (min) 3,393 mg. [2]Amino acids guaranteed percentage of purity: L-Lysine = 99%; L-Threonine = 98.5%; DL-Methionine = 99%.
Source: Elaborated by the authors (2024).

Table 2. Nutritional composition of the feed provided to the swine in the finishing stage

Ingredients (%)	Levels of WV (%)			
	0.0	0.5	1.0	1.5
Cornmeal 7.88%	75.31	75.31	75.31	75.31
Soybean meal 45%	19.39	19.39	19.39	19.39
commercial supplement (evimix)	2.50	2.50	2.50	2.50
Soy oil	1.00	1.00	1.00	1.00
L-Lysine[2]	0.22	0.22	0.22	0.22
L-Threonine[2]	0.050	0.050	0.050	0.050
DL-Methionine[2]	0.024	0.024	0.024	0.024
Wood vinegar	0.00	0.50	1.00	1.50
Sand	1.50	1.00	0.50	0.00
Total	100.00	100.00	100.00	100.00
Calculated composition				

Metabolizable energy (kcal kg^{-1})	3,235	3,235	3,235	3,235
Crude protein (%)	15.74	15.74	15.74	15.74
Digestible lysine (%)	0.814	0.814	0.814	0.814
Digestible methionine (%)	0.252	0.252	0.252	0.252
Digestible threonine (%)	0.545	0.545	0.545	0.545
Digestible tryptophan (%)	0.147	0.147	0.147	0.147

[1]Guaranteed levels per kg of product: folic acid (min) 5 mg; panthotenic acid (min) 257 mg; BHT (min) 133 mg; bacitracin (min) 1.333.33 mg; biotin (min) 3.0 mg; calcium (min) 230 g; calcium (max) 240 g; cobalt (min) 5.66 mg; copper (min) 4,000 mg; choline (min) 3,330 mg; iron (min) 3,333 mg; phytase (min) 16.65 ftu; fluorine (max) 332 mg; phosphorous (min) 34.6 g; iodine (min) 33.33 mg; manganese (min) 1,333 mg; niacin (min)569 mg; selenium (min) 10 mg; sodium (min) 58.5 g; vitamin A (min) 116,800 UI; vitamin B1 (min) 16.7 mg; vitamin B12 (min) 353 mg; vitamin B2 (min) 66.7 mg; vitamin B6 (min) 16.7 mg; vitamin D3 (min) 25,000 UI; vitamin C (min) 833.33 UI; vitamin K3 (min) 49 mg; zinc (min) 3.393 mg. [2]Amino acids guaranteed percentage of purity: L-Lysine = 99%; L-Threonine = 98.5%; DL-Methionine = 99%.
Source: Elaborated by the authors (2024)

Eucalyptus WV was obtained from wood carbonization from a hybrid of *Eucalyptus urophylla* x *Eucalyptus grandis* (*Eucalyptus urograndis* – clone GG 100). All the procedures for producing and purifying the eucalyptus WV and its chemical composition were described previously in detail (Pimenta et al. 2018). Feijó et al. (2022) evaluated EP as an antiseptic in post-surgical procedures in cats and observed inhibition of the action of gram-positive and gram-negative bacteria. Chu et al. (2013) analyzed the performance, immune response, and population of the fecal microflora of growing pigs fed with charcoal diets with bamboo WV as alternatives to antibiotics. The same research work reported the chemical profile of the WV, including the composition of the phenolic fraction, which contains compounds with biological effects such as antibiotic action (Araújo et al. 2017; Pimenta et al. 2018). Table 3 displays the main compounds with biological activity annotated by the authors in the eucalyptus WV's total ion chromatogram.

Table 3. Main compounds with biological activity annotated in the total ion chromatogram of the eucalyptus WV

Compound	Molecular formula	Similarity (%)	Area (%)
Furfural	$C_5H_4O_2$	99	15.67
2-methoxy-phenol (guaiacol)	$C_7H_8O_2$	98	16.31
3-methyl-2-methoxy-phenol	$C_8H_{10}O_2$	96	0.92
2,6-dimethyl-phenol	$C_8H_{10}O$	97	0.19
2-methoxy-5-methyl-phenol	$C_8H_{10}O_2$	97	0.63
4-methyl-2-methoxy-phenol	$C_8H_{10}O_2$	97	4.87
Phenol	C_6H_6O	98	1.43
2-methyl-phenol (o-cresol)	C_7H_8O	97	1.21
4-ethyl-2-methoxy-phenol	$C_9H_{12}O_2$	98	2.61
4-methyl-phenol (p-cresol)	C_7H_8O	97	0.67
2,6-dimethyl-phenol (2,6-xylenol)	$C_8H_{10}O$	95	0.31
3-methyl-phenol (m-cresol)	C_7H_8O	96	0.49
2,5-dimethyl-phenol (2,5-xylenol)	$C_8H_{10}O$	87	0.58
3,4-dimethoxy-phenol	$C_8H_{10}O_3$	91	0.17
4-propyl-2-methoxy-phenol	$C_{10}H_{14}O_2$	92	0.37
2,4-dimethyl-phenol (2,4-xylenol)	$C_8H_{10}O$	85	0.14
3-allyl-6-methoxy-phenol	$C_{10}H_{12}O_2$	97	0.15
2,6-dimethoxy-phenol (syringol)	$C_8H_{10}O_3$	97	8.48

4-methyl-2,6-dimethoxy-phenol	$C_9H_{12}O_3$	80	2.52
2,6-dimethoxy-4-allyl-phenol	$C_{11}H_{14}O_3$	88	0.11
TOTAL			57.83
OTHER COMPOUNDS			42.17

Source: Araujo et al. 2017, Pimenta et al. 2018

The animals received feed and water at will. The feed leftovers were weighed daily to determine the daily feed consumption (DFC). The animals were weighed at the beginning and end of the experiment to calculate the daily weight gain (DWG) and feed conversion. The score of the animals' feces was evaluated throughout the experimental period, according to the method described by Vassalo et al. (1997), where a number is assigned to each state of the feces: 1 represents feces with regular consistency, 2 denotes soft stools, and 3 indicates diarrhea. At the end of the experimental period, when the animals reached an average weight of 97.5 + 5.1 kg, they were subjected to dry fasting for 8 hours, left to rest for a further 4 hours, and then slaughtered for a total of 12 hours of fasting.

Slaughter was performed through stunning by electronarcosis, followed by bleeding by perforation of the jugular vein. The trimming and evisceration were carried out, and the carcasses were sawn longitudinally in half along the back-lumbar line, followed by weighing to obtain the hot carcass weight. Then, all the parts were cooled in a cold chamber at four ºC for 24 hours to evaluate the following quantitative parameters: carcass length, backfat thickness, backfat thickness (P2), cold carcass weight, ham weight, carcass yield, meat yield, meat yield in the cooled carcass, weight loss on cooling, loin depth, loin eye area, fat area, and meat: fat ratio. The pH, marbling, and muscle color were evaluated to analyze the qualitative parameters. The carcass length was measured from the cranial edge of the pubic symphysis to the ventral cranial edge of the atlas using a tape measure according to the Brazilian Method of Carcass Classification (ABCS, 1973).

The half carcasses were individually weighed, and the left half carcass was cut at point P2 (corresponding to the perpendicular projection of the last rib at 4 cm from the spine) to expose the longissimus dorsi muscle and the backfat layer. To measure the backfat thickness and the depth of the loin (longissimus dorsi), the technique described by ABCS (1973) was followed. The backfat thickness was measured at three points on the carcass: at the height of the first rib (BT1), on the last rib (BT2), and between the previous and penultimate lumbar spine (BT3), with the aid of a caliper. Subsequently, the average backfat thickness (ABT) was calculated based on the three measurements. The measurement of the loin eye area (longissimus dorsi) was performed at the height of the last rib (in the region of insertion of the last thoracic vertebra with the first lumbar), outlining the muscle area on tracing paper, which was then measured with the aid of a planimeter.

The ham cut was obtained by sectioning the joint between the last lumbar vertebra, perpendicular to the dorsal line of the carcass. According to the procedures described in the ABCS

recommendations (ABCS 1973), the ham must be weighed completely, with tail and paw without a nail and any touches on the meat and fat. The carcasses were weighed again after 24 hours in the cold chamber when the muscle was converted into meat to obtain the weight of the cold carcass. With these values, the carcass loss during cooling was estimated using Equation 1.

Equation 1:

$$CL\ (\%) = \left[\frac{\text{HCW} - \text{CCW}}{\text{CCW}}\right] x\ 100$$

Where:
CL = carcass loss (%)
HCW = hot carcass weight (kg)
CCW = cold carcass weight (kg)

The measurement of muscle depth was performed according to the technique described by Bridi and Silva (2009). It was measured at the height of the last rib, in the region of insertion of the last thoracic vertebra with the first lumbar spine, 6 cm from the midline of the carcass cut (point P2). The values obtained using a caliper were expressed in millimeters. For the correct assessment of the depth of the longissimus dorsi muscle, the caliper was oriented from point P2 perpendicularly to the extreme opposite end of the muscle.

To obtain the carcass yield, Equation 2 was employed.

Equation 2:

$$CY\ (\%) = \left[\frac{\text{HCW}}{\text{LW}}\right] x\ 100$$

Where:
CY = carcass yield (%)
HCW = hot carcass weight (kg)
LW = live weight (kg)

The meat yield was calculated using Equation 3, as proposed by Irgang (1998).

Equation 3:

$$MY\ (\%) = 60 - (ET2\ x\ 0.58) + (MD\ x\ 0.10)$$

Where:
MY = meat yield (%)
ET2 = backfat thickness at the last rib (mm)
MD = muscle depth (mm)

The amount of meat in the carcass was calculated according to Equation 4.

Equation 4:

$$AM = \frac{(\text{CCW x MY})}{100}$$

Where:
AM = amount of meat in the carcass (kg)
CCW = cold carcass weight (kg)
MY = meat yield (%)

Equation 5, proposed by Guidoni (2000), was used to calculate the meat yield of the cold carcass.

Equation 5:

$$MYCC\ (\%) = 65.92 - [(0.685\ x\ ET2) + (0.094\ x\ MD) - (0.026\ x\ CCW)]$$

Where:
MYCC = meat yield of the cold carcass (%)
CCW = cold carcass weight (kg)
MY = meat yield (%)
ET2 = backfat thickness at the last rib (mm)
MD = muscle depth (mm)

To determine the loin eye area in the longissimus dorsi muscle and fat area (in the backfat), a perpendicular cut was made at the height of the last rib (in the region of insertion of the last thoracic vertebra with the first lumbar spine) to expose the muscle and fat. The outline was drawn on tracing paper with a permanent ink pen with the appropriate rib marking, which was later measured with a

planimeter, according to the method described by ABCS (1973). With the longissimus dorsi muscle and fat values, the meat/fat ratio was calculated according to Equation 6.

Equation 6:

$$MFR = \frac{\text{LEA}}{\text{FA}}$$

MFR = meat/fat ratio
LEA = loin eye area (mm^2)
FA = fat area (mm^2)

The temperature and the pH were measured 45 min and 24 hours after slaughter. The pH was determined using a portable pH meter (Hl 99163 – Hanna Instruments). The color of the muscle was determined by employing a color chart with a 1 to 6 scale (National Pork Producers Council, 1999), where 1 and 6 correspond to meat with low and high pigmentation, respectively. The marbling was scored on a scale ranging from 1 to 7, where the lowest and the highest values correspond to only traces of marbling and excessive marbling, respectively (AMSA – American Meat Science Association, 2001).

The statistical analysis of the experimental data was performed using the routine GLM of the software SAS 9.0 (Statistical Analysis System, SAS Institute), decomposing the sum of the squares of treatments in orthogonal contrasts. The data were subjected to variance analysis. The statistical models were fitted by fixing the effect of the eucalyptus WV and the experimental block. Regression models analyzed the impact of treatments. The linear and quadratic models were tested, and the best models were selected based on the following decision criteria: coefficient of correlation (r) between experimental and estimated values, the significance of the parameters in the equation, biological realism of the model, Akaike information criterion, and root-mean-square error (RMSE). In general, a correlation coefficient nearer to 1 and lower RMSE indicate better estimation power of the model, adopting the level of 95% probability. The following statistical model was employed:

$$Y_i = B_0 + \beta_1 + X_i + \beta_2 X_i^2 + e_i$$

Where:
Y_i = observed value of the dependent variable
B_0 = regression constant
β_1 = regression coefficient
X_i = level of the independent variable eucalyptus WV (0, 0.5, 1.0, and 1.5%)
β_2 = coefficient of regression
X_i = level of the squared independent variable eucalyptus WV
e_i = error associated with the observed values.

3 RESULTS AND DISCUSSION

An increasing linear trend was observed for the effect of rising levels of WV on the variable daily feed consumption (DFC) and daily weight gain (DWG), according to the equations displayed in Figures 1 and 2. However, there were no effects on the feed conversion (FC) and fecal score (Table 4).

Table 4. Performance and score of the feces from the swine

Variables	WV levels (%)				CV	P-value
	0.0	0.5	1.0	1.5		
Daily feed consumption *	2.543	2.603	2.823	2.868	7.04	0.003
Daily weight gain *	0.876	0.900	0.966	0.963	8.19	0.032
Feed conversion ^{ns}	2.90	2.89	2.92	2.98	7.85	0.489
Analysis of feces						
Score of the feces ^{ns}	1.123	1.213	1.208	1.225	14.03	0.337

CV = coefficient of variation; *Linear effect observed; ns = no significant effect
Source: Elaborated by the authors (2024).

The experimental results of the DFC suggest that the eucalyptus WV stimulated the consumption of feed by the animals, promoting an effect similar to a palatabilizer. This sort of effect most likely can be explained by the presence of some components in the WV, such as furfural, syringol, and guaiacol, which have organoleptic properties and make the feed more appealing to the animals, improving the consumption and, therefore, the weight gain. Nevertheless, the results achieved in the experiment for the DFC and weight gain were insufficient to affect the feed conversion variable.

The effect of conventional antibiotics or natural products on animal performance is related to their respective action on the intestinal microflora. Reducing the concentration of pathogenic bacteria in the intestinal tract improves nutrient digestibility by decreasing the inflammatory responses to subclinical infections (BLANK et al., 2001, DIBNER & BUTTIN, 2002), increasing nutrient absorption. When assessing the effect of rising concentrations of wood vinegar (0, 0.2, 0.4, and 0.8%) and antibiotics (aureomycin, colistin sulfate, kitasamycin at 75, 20, and 50 mg kg 1, respectively) in weanling pigs, WANG et al. (2012) observed at the end of the experimental period of 25 days that the group that received 0.4% WV in the diet and the ones that received antibiotic obtained similar results. However, the feed conversion determined in the cited experiment was lower than our observed values. Even though in this study, the feed conversion did not present any statistically significant behavior due to the action of WV, a daily weight gain of 10% was observed, a percentage that can generate substantial savings in swine farming due to the shorter slaughter time. In other words, considering the slaughter weight and timing, it would be possible to reduce this period by 10%, meaning 36 days a year would be saved, correspondingly reducing all operating expenses (water, feed, energy, and labor).

The animals' feces were evaluated daily to measure whether the WV positively affected the diarrhea index, a problem that can particularly affect piglets after leaving the nursery. However, the results showed no influence of the product on the appearance of the feces. In the present study, no diarrhea was observed in the animals. However, this condition is more common in weaned piglets, whether due to changes in the composition of the diet or bacterial infections; all stages of development are subject to this problem. Other authors, such as *Van Chao* (2016) and Kupittayanant and Kupittayanant (2017), evaluated the addition of 5% WV compared to enrofloxacin on the indexes of diarrhea in wealing pigs and observed similar effects to those from the ministration of antibiotics and concluded that the inclusion of WV not only significantly reduced the count of Escherichia coli but also increased the population of Lactobacillus spp., bringing benefits to the animals.

The carcass characteristics were assessed, but no significant effects were detected for the quantitative variables (Table 5): backfat thickness (BT1, BT2, and BT3); average backfat thickness (ABT); carcass length (CL); hot carcass weight (HCW); cold carcass weight (CCW); ham weight (HW); carcass yield (CY); backfat thickness (P2); meat yield from the cold carcass (MYCC); meat yield, meat amount from the carcass (MAC); loin depth (LD); loin eye area (LEA); fat area (FA); and meat: fat ratio (MFR).

Table 5. Quantitative variables obtained for the swine carcasses

Variables	Levels of WV (%)				CV	P-value
	0.0	0.5	1.0	1.5		
Backfat thickness[1]	3.06	3.26	3.46	3.27	9.77	0.170
Backfat thickness[2]	2.06	2.89	2.50	2.13	24.38	0.914
Backfat thickness[3]	2.14	1.94	2.41	1.96	30.80	0.961
Average backfat thickness	2,42	2.60	2.79	2.45	16.71	0.715
Carcass length	100.16	97.66	98.33	98.50	2.71	0.385
Hot carcass weight	77.10	78.75	82.23	80.16	7.43	0.253
Cold carcass weight	75.81	76.13	80.35	79.56	7.34	0.154
Ham weight	11.59	11.82	12.54	12.15	8.69	0.225
Carcass yield	81.96	81.83	81.80	81.67	1.83	0.748
Meat yield in the cold carcass	63.54	63.43	63.13	63.46	0.58	0.430
Meat yield	60.57	60.54	60.52	60.56	0.10	0.602
Amount of meat in the carcass	45.92	46.09	48.63	48.18	7.37	0.157
Backfat thickness P2	1.50	1.61	1.88	1.45	26.90	0.884
Loin depth	6.63	6.41	6.36	6.45	7.32	0.495
Area of loin eye	41.67	41.94	40.49	39.25	9.92	0.251
Fat area	16.24	19.55	20.46	20.09	28.16	0.218
Meat: fat ratio	2.28	2.35	2.03	1.99	25.08	0.243

CV = coefficient of variation
Source: Elaborated by the authors (2024).

Regarding the carcass characteristics, studies assessing the influence of WV are scarce. The positive effects of WV's components on the animals' performance and the negative impact that can be caused should be evaluated. The absence of effect on the carcass parameters can be positive since no harm was done by including the WV. Kook et al. (2005) studied the addition of 5% bamboo wood vinegar (2 and 4%) in the swine diet on the characteristics of carcasses of animals in the finishing stage. The authors observed a significant increase in the final carcass weight by adding 4% bamboo WV. Also, the dorsal fat thickness decreased by using 2 and 4% of this type of WV. However, they reported an increase in the total loin fat of the animals that received the bamboo WV. The authors also assessed the meat quality and observed that yellowing was significantly higher than in the control group, while the redness decreased. In the sensory analysis, the authors detected differences among the experimental treatments, with a decrease in the odor and an improvement in the appearance of the meat to consumers. According to Ramos and Gomide (2007), color is one of the leading influencers of perceived meat quality, directly affecting consumers' acceptance of the product. Moreover, no significant differences were detected for the qualitative variables (Table 6): temperature, pH, color, and marbling.

Table 6. Qualitative variables of the swine carcasses

Variables	Levels of WV (%)				CV	P-value
	0.0	0.5	1.0	1.5		
pH 45	6.57	6.71	6.40	6.79	4.96	0.567
T º 45	30.33	30.23	31.30	30.93	4.57	0.276
pH 24	5.75	5.77	5.57	5.86	4.82	0.765
T º 24	8.85	8.68	8.28	8.83	10.56	0.790
Color	3.00	3.00	3.00	2.67	17.58	0.297
Marbling	2.50	2.17	2.67	2.33	36.86	1.000

CV = coefficient of variation
Source: Elaborated by the authors (2024).

The marbling was not affected by the inclusion of rising levels of WV in the diet so that property was acceptable to guarantee tasty and juicy meat. According to Bridi and Silva (2009), fat levels under 2% negatively affect meat quality since intramuscular fat is important in conserving meat, besides improving its juiciness, taste, and tenderness. According to Ferreira et al. (2011), the marbling index of pork can range from 1 to 3%, where values below 1% denote meat with a deficit of juiciness and palatability. In contrast, values above 3% represent acceptable qualities. In the present work, a marbling index higher than 2% was observed, which is in the range recommended in the literature. Thus, the levels of WV employed in the experiment were not harmful to the meat quality parameters considered.

4 CONCLUSIONS

The use of wood vinegar as an additive in pig diets in the growing and finishing phases provided greater growth and development and reduced the time taken for animals to arrive at slaughter, potentially generating savings in labor, water, and electrical energy. Therefore, it is recommended to use up to a level of 1.5%; however, new research must be developed to evaluate higher levels.

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